4

1

2

1

SIR (Susceptible-Infectious-Recovered).

2

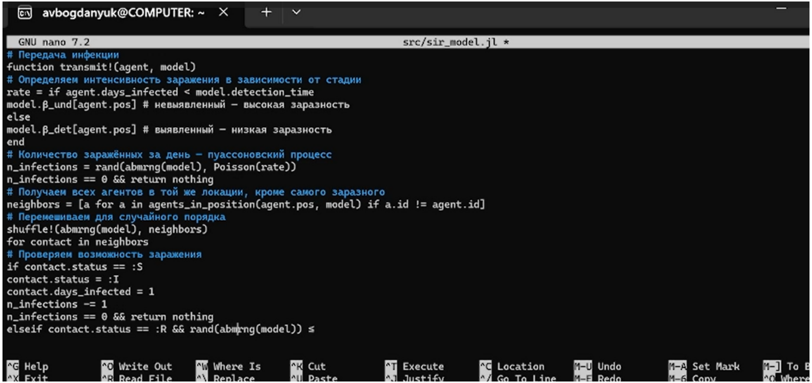
SIR, 1927 , , : —
(Susceptible) — ; — (Infectious) — , ;
— (Recovered) — ( ), .

— , : — status::Symbol — (:S, :I, :R); —
days_infected::Int — ( )

— Ns — ; — _und — ; — _det —
; — infection_period — ( ); — detection_time
— ; — death_rate — ; — reinfection_probability —
; — migration_rates — .

: — — —
— , . — —
days_infected. — — days_infected
infection_period, (R), ( ) death_rate.
reinfection_probability.

. , (. 1).

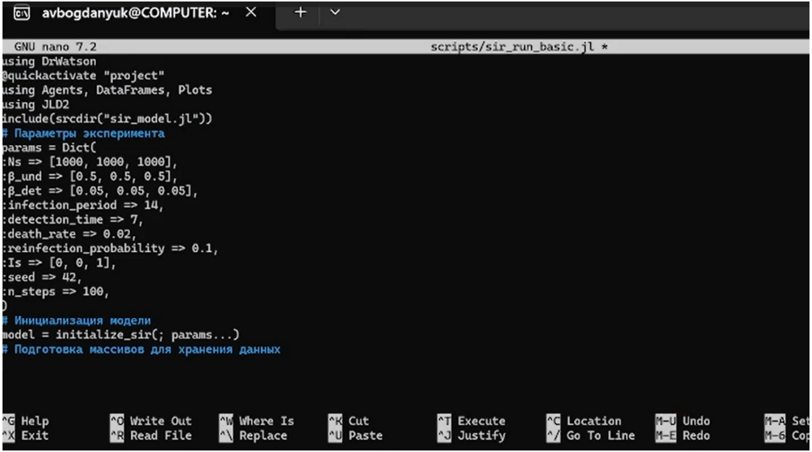


```
GNU nano 7.2                                    src/sir_model.jl *
# Передача инфекции
function transmit!(agent, model)
# Определяем интенсивность заражения в зависимости от стадии
rate = if agent.days_infected < model.detection_time
model.β_und[agent.pos] # невыявленный – высокая заразность
else
model.β_det[agent.pos] # выявленный – низкая заразность
end
# Количество заражённых за день – пуассоновский процесс
n_infections = rand(abmrng(model), Poisson(rate))
n_infections == 0 && return nothing
# Получаем всех агентов в той же локации, кроме самого заразного
neighbors = [a for a in agents_in_position(agent.pos, model) if a.id != agent.id]
# Перемешиваем для случайного порядка
shuffle!(abmrng(model), neighbors)
for contact in neighbors
# Проверяем возможность заражения
if contact.status == :S
contact.status = :I
contact.days_infected = 1
n_infections -= 1
n_infections == 0 && return nothing
elseif contact.status == :R && rand(abmrng(model)) ≤
```

1: src/sir_model.jl

( ) .

. (. 2).



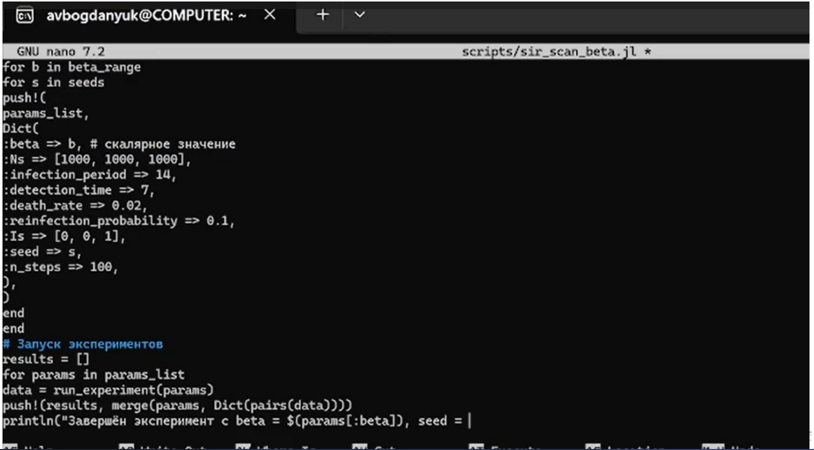
```
avbogdanyuk@COMPUTER: ~
GNU nano 7.2                                   scripts/sir_run_basic.jl *
using DrWatson
@quickactivate "project"
using Agents, DataFrames, Plots
using JLD2
include(srcdir("sir_model.jl"))
# Параметры эксперимента
params = Dict(
:Ns => [1000, 1000, 1000],
:β_und => [0.5, 0.5, 0.5],
:β_det => [0.05, 0.05, 0.05],
:infection_period => 14,
:detection_time => 7,
:death_rate => 0.02,
:reinfection_probability => 0.1,
:Is => [0, 0, 1],
:seed => 42,
:n_steps => 100,
)
# Инициализация модели
model = initialize_sir(; params...)
# Подготовка массивов для хранения данных

^G Help      ^O Write Out  ^W Where Is  ^K Cut    ^T Execute  ^C Location   M-U Undo  M-A Set
^X Exit      ^R Read File  ^\ Replace   ^U Paste  ^J Justify  ^/ Go To Line M-E Redo  M-6 Cop
```

2: src/sir_run_basic.jl

, , (_und _det)

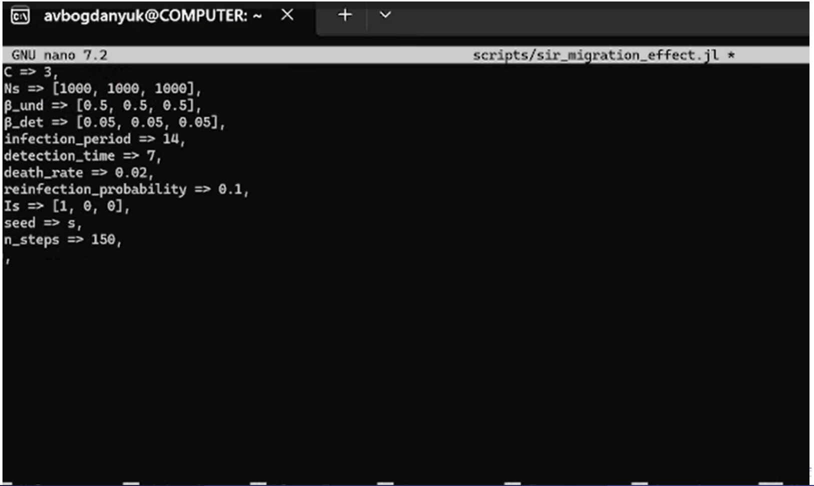
: , , .

(. 3).

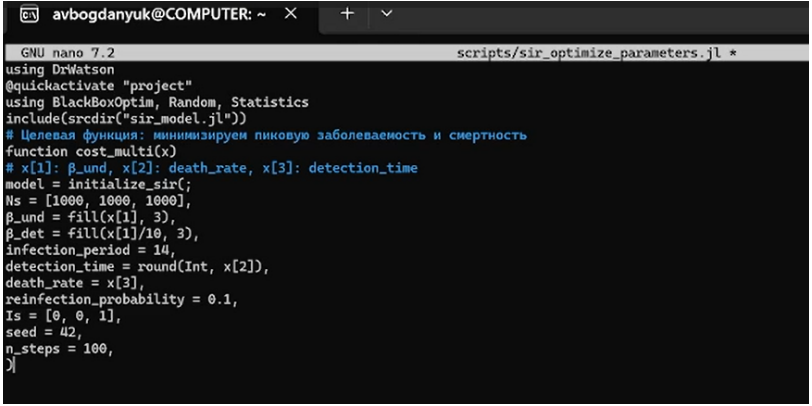


```
GNU nano 7.2                                  scripts/sir_scan_beta.jl *
for b in beta_range
for s in seeds
push!(
params_list,
Dict(
:beta => b, # скалярное значение
:Ns => [1000, 1000, 1000],
:infection_period => 14,
:detection_time => 7,
:death_rate => 0.02,
:reinfection_probability => 0.1,
:Is => [0, 0, 1],
:seed => s,
:n_steps => 100,
),
)
end
end
# Запуск экспериментов
results = []
for params in params_list
data = run_experiment(params)
push!(results, merge(params, Dict(pairs(data))))
println("Завершён эксперимент с beta = $(params[:beta]), seed = 
```

, ,   (
)   .   ,   (. 4).

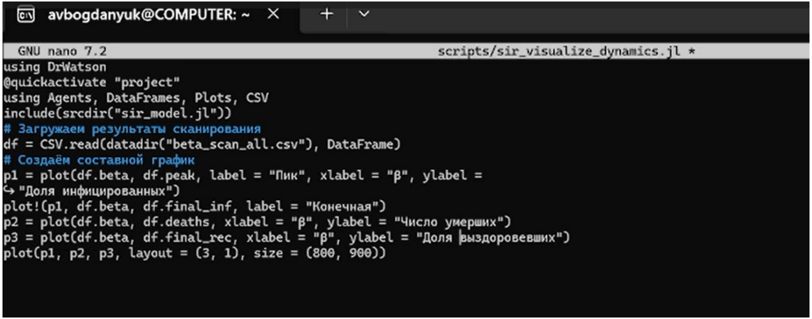

avbogdanyuk@COMPUTER: ~
GNU nano 7.2
scripts/sir_migration_effect.jl *
C => 3,
Ns => [1000, 1000, 1000],
β_und => [0.5, 0.5, 0.5],
β_det => [0.05, 0.05, 0.05],
infection_period => 14,
detection_time => 7,
death_rate => 0.02,
reinfection_probability => 0.1,
Is => [1, 0, 0],
seed => s,
n_steps => 150,
),

, , :
. (Borg MOEA) BlackBoxOptim (. 5).



5: src/sir_optimize_parameters.jl

, (data/beta_scan_all.csv, scan_beta.jl) , : , , (. 6).

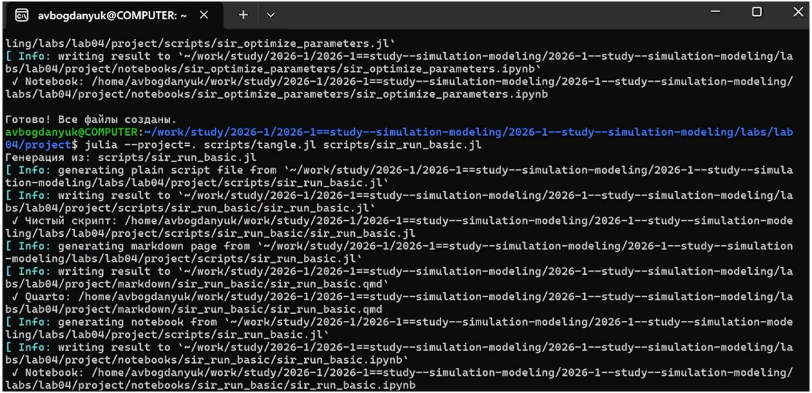


```
GNU nano 7.2                                   scripts/sir_visualize_dynamics.jl *
using DrWatson
@quickactivate "project"
using Agents, DataFrames, Plots, CSV
include(srcdir("sir_model.jl"))
# Загружаем результаты сканирования
df = CSV.read(datadir("beta_scan_all.csv"), DataFrame)
# Создаём составной график
p1 = plot(df.beta, df.peak, label = "Пик", xlabel = "β", ylabel =
↪ "Доля инфицированных")
plot!(p1, df.beta, df.final_inf, label = "Конечная")
p2 = plot(df.beta, df.deaths, xlabel = "β", ylabel = "Число умерших")
p3 = plot(df.beta, df.final_rec, xlabel = "β", ylabel = "Доля выздоровевших")
plot(p1, p2, p3, layout = (3, 1), size = (800, 900))
```

6: src/sir_visualize_dynamics.jl

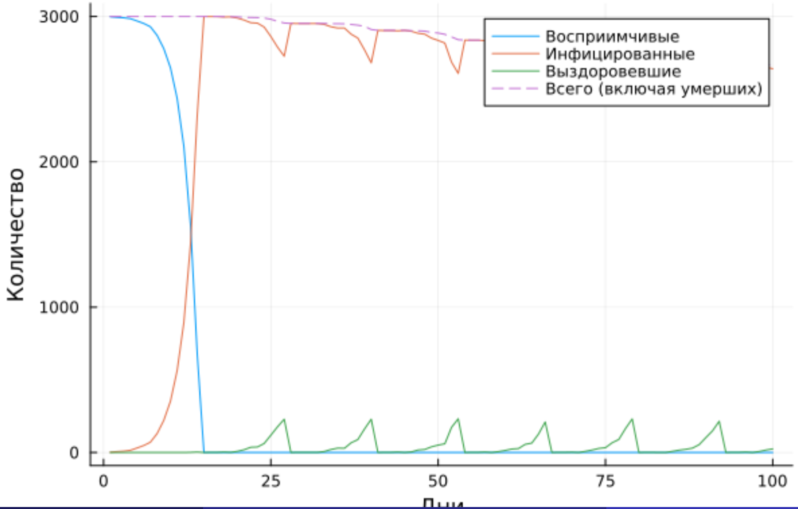
, , . tangle.jl

(. 7).

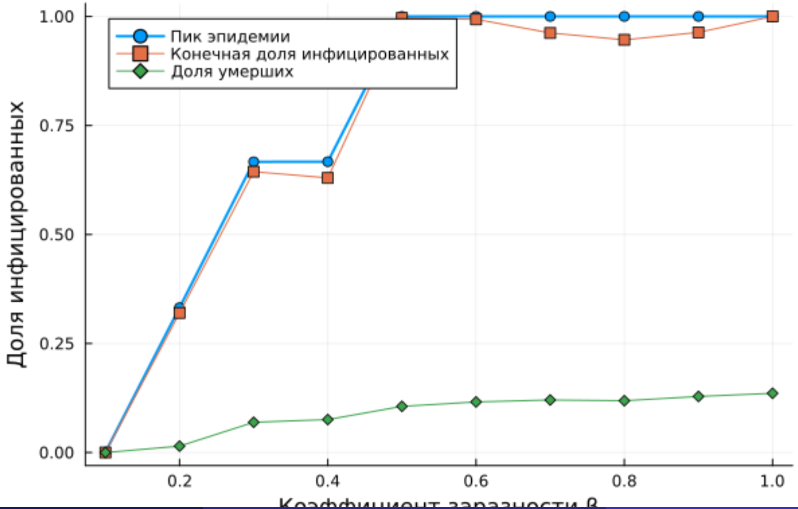

avbogdanyuk@COMPUTER: ~
ling/labs/lab04/project/scripts/sir_optimize_parameters.jl`
[Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/notebooks/sir_optimize_parameters/sir_optimize_parameters.ipynb`
✓ Notebook: /home/avbogdanyuk/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/notebooks/sir_optimize_parameters/sir_optimize_parameters.ipynb
Готово! Все файлы созданы.
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project$ julia --project=. scripts/tangle.jl scripts/sir_run_basic.jl
Генерация из: scripts/sir_run_basic.jl
[Info: generating plain script file from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/scripts/sir_run_basic.jl`
[Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/scripts/sir_run_basic/sir_run_basic.jl`
✓ Чистый скрипт: /home/avbogdanyuk/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/scripts/sir_run_basic/sir_run_basic.jl
[Info: generating markdown page from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/scripts/sir_run_basic.jl`
[Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/markdown/sir_run_basic/sir_run_basic.qmd`
✓ Quarto: /home/avbogdanyuk/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/markdown/sir_run_basic/sir_run_basic.qmd
[Info: generating notebook from `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/scripts/sir_run_basic.jl`
[Info: writing result to `~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/notebooks/sir_run_basic/sir_run_basic.ipynb`
✓ Notebook: /home/avbogdanyuk/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab04/project/notebooks/sir_run_basic/sir_run_basic.ipynb

7:

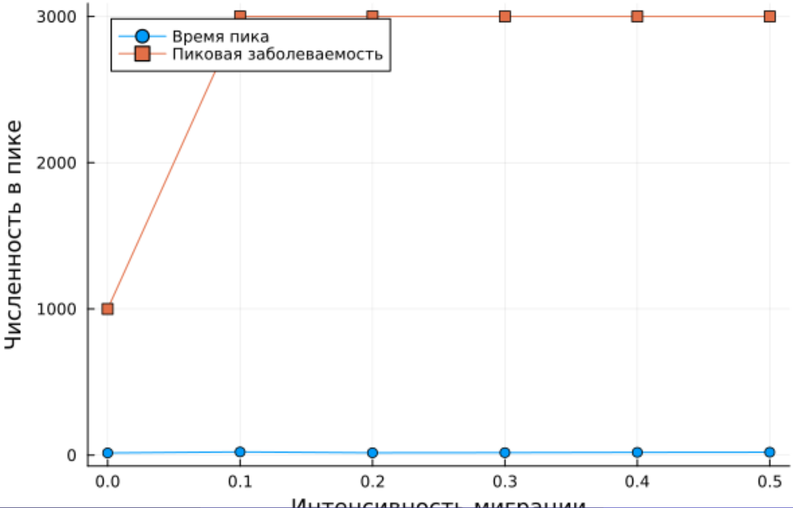
sir_run_basic.jl (. 8).



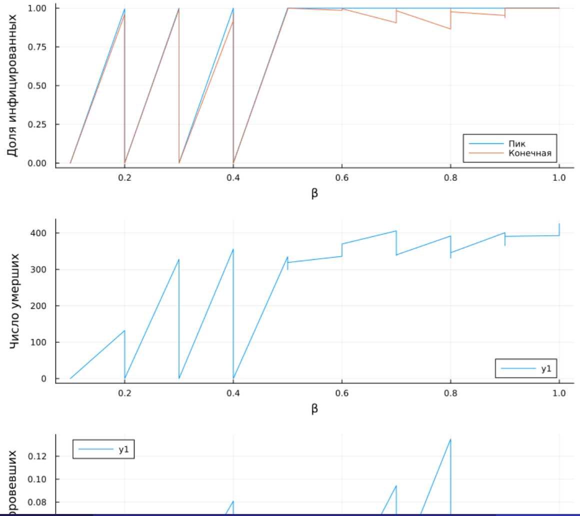
sir_beta_scan.jl (. 9).



sir_migration_effect.jl (. 10).



sir_visualize_dynamics.jl (. 11).



SIR (Susceptible-Infectious-Recovered).

1. Hethcote H. W. The Mathematics of Infectious Diseases // SIAM Review. — 2000. — Jan. — Vol. 42, no. 4. — P. 599–653. — ISSN 1095-7200. — DOI: 10.1137/s0036144500371907.

1. Hethcote H. W. The Mathematics of Infectious Diseases // SIAM Review. — 2000. — Jan. — Vol. 42, no. 4. — P. 599–653. — ISSN 1095-7200. — DOI: 10.1137/s0036144500371907.
2. Kermack W. O., McKendrick A. G. A Contribution to the Mathematical Theory of Epidemics // Proceedings of the Royal Society of London. Series A Containing Papers of a Mathematical and Physical Character. — 1927. — . — . 115, № 772. — . 700—721. — ISSN 2053-9150. — DOI: 10.1098/rspa.1927.0118.